

Topic C5

Chemical energetics

Concept guide · Worked examples · Cambridge-style practice · Fully explained answers

What this guide teaches

A complete explanation of the syllabus concepts, chemical reasoning, worked applications, common misconceptions and original exam-style practice for this topic.

Syllabus map

C5.1	Exothermic and endothermic reactions
C5.2	Reaction pathway diagrams
C5.3	Bond energies
C5.4	Energy sources

How to study: retrieve definitions and equations, cover the worked examples, then complete the practice paper without using the notes.

1. Energy transfer

Exothermic reactions transfer thermal energy to the surroundings, causing temperature to rise; products have lower chemical energy than reactants. Endothermic reactions take in thermal energy, causing temperature to fall; products have higher chemical energy. Combustion and neutralisation are exothermic; thermal decomposition and photosynthesis are endothermic.

$\Delta H < 0$ exothermic $\Delta H > 0$ endothermic

Examiner insight

State the direction of energy transfer relative to the reacting system, not simply “heat is produced”.

2. Reaction pathway diagrams

A pathway diagram plots energy against progress of reaction. Activation energy is the minimum energy colliding particles need for a successful reaction. The energy change is product energy minus reactant energy. A catalyst provides an alternative pathway with lower activation energy but does not change reactant energy, product energy or overall energy change.

Examiner insight

Label both energy levels and draw activation energy from the reactant level to the peak.

3. Bond breaking and bond making

Breaking bonds is endothermic because energy is needed to overcome attractions. Making bonds is exothermic because energy is released. A reaction is exothermic when energy released by forming product bonds exceeds energy absorbed breaking reactant bonds.

$\Delta H = \text{energy to break bonds} - \text{energy released making bonds}$

Worked example

For $\text{H}_2 + \text{Cl}_2 \rightarrow 2\text{HCl}$: break $436+242$, form 2×431 ; $\Delta H = 678-862 = -184 \text{ kJ/mol}$.

Examiner insight

Count every bond shown in the balanced equation, including coefficients.

4. Experimental energy changes

A simple calorimetry experiment measures temperature change of a known mass or volume of solution. Improve validity with insulation, a lid and stirring. Heat loss makes measured exothermic changes less negative. Fair comparisons require same volumes, concentrations and initial temperatures.

energy transferred = mass $\times$ specific heat capacity $\times$ temperature change

Examiner insight

Distinguish repeatability (same method/operator) from accuracy (closeness to true value).

5. Fuels and energy resources

A fuel releases energy when it reacts, usually by combustion. Compare fuels using energy per gram, cost, availability, ease of storage and pollutants. Hydrogen forms water at use but its production route matters. Fossil fuels are finite and emit carbon dioxide; biofuels may be renewable but use land and resources.

Examiner insight

Environmental evaluation needs advantages, disadvantages and a justified conclusion linked to evidence.

Exam technique

High-value habits

Use precise chemical vocabulary, balance equations, show every calculation step, include units and state symbols where required, distinguish observations from conclusions, and answer the command word directly.

Retrieval checklist

- Recall all definitions, tests and equations without notes.
- Explain one unfamiliar observation using particles, bonding or collision theory.
- Check charge, atoms, units and significant figures in every calculation.
- Complete one timed set and mark it only after finishing.

Practice Paper

Topic C5

hemical energetics

Instructions

Answer all questions. Select one option for multiple choice. Show working in open spaces and write explanations on the lines. Individual parts carry 1-3 marks.

Candidate name:

Class / date:

Total marks: 50

Suggested time: 60 minutes.

Section A · Multiple choice

Choose the best answer for each question.

1. Which change is exothermic? [1]
 - A photosynthesis
 - B thermal decomposition
 - C combustion
 - D bond breaking only

2. In an endothermic reaction, products have [1]
 - A less energy than reactants
 - B more energy than reactants
 - C zero energy
 - D the same energy always

3. Activation energy is the [1]
 - A total energy in products
 - B minimum energy for successful collision
 - C energy released by cooling
 - D temperature change

4. A catalyst changes the [1]
 - A overall enthalpy change
 - B product energy
 - C activation energy
 - D reactant formula

5. Breaking chemical bonds is [1]
 - A always exothermic
 - B always endothermic
 - C energy neutral
 - D a catalyst

6. Making bonds [1]
 - A absorbs energy
 - B releases energy
 - C removes atoms
 - D changes proton number

7. If bond breaking needs 500 kJ and bond making releases 650 kJ, Delta H is [1]
 - A +1150
 - B +150
 - C -150
 - D -1150

8. Which reduces heat loss in calorimetry? [1]
- A metal lid removed
 - B polystyrene cup and lid
 - C larger open flame
 - D no insulation
9. Why may experimental combustion values be too low? [1]
- A Complete insulation
 - B heat loss and incomplete combustion
 - C extra product bonds
 - D mass cannot be measured
10. Which fuel forms only water when completely used in a fuel cell? [1]
- A coal
 - B methane
 - C hydrogen
 - D ethanol

Section B · Structured questions

11 A reaction mixture becomes warmer.

(a) State whether the reaction is exothermic or endothermic. [1]

(b) Describe the direction of thermal energy transfer. [2]

(c) Compare energy of products and reactants. [2]

(d) Give two examples of this type of reaction and one observation other than temperature. [3]

[Total: 8]

12 A pathway diagram represents an exothermic reaction.

(a) Describe the relative positions of reactants and products. [2]

(b) Define activation energy. [2]

(c) Describe the effect of a catalyst on the curve. [2]

(d) State two quantities unchanged by the catalyst. [2]

[Total: 8]

13 Use bond energies for $\text{H}_2 + \text{Br}_2 \rightarrow 2\text{HBr}$: H-H 436, Br-Br 193, H-Br 366 kJ/mol.

(a) Calculate energy required to break bonds. [2]

(b) Calculate energy released making bonds. [2]

(c) Calculate Delta H. [2]

(d) Classify and explain the reaction. [2]

[Total: 8]

14 A student compares two fuels by heating water.

(a) State two measurements required. [2]

(b) State two control variables. [2]

(c) Suggest two improvements to reduce energy loss. [2]

(d) Explain why incomplete combustion affects the result. [2]

[Total: 8]

15 Evaluate hydrogen and methane as fuels.

(a) State the complete-combustion products of methane. [2]

(b) State the product from hydrogen. [1]

(c) Give two advantages of hydrogen at point of use. [2]

(d) Give two limitations and a justified conclusion. [3]

[Total: 8]

Worked solutions

Remove or fold back this section while completing the practice paper.

Marking routine

Mark each part separately. Award method and chemistry marks where shown, then identify the first point at which an answer becomes incorrect.

Solutions · Multiple choice

- 1 C. Combustion transfers energy to surroundings.
- 2 B. The system has absorbed energy.
- 3 B. It is the threshold for reaction.
- 4 C. It supplies a lower-energy pathway.
- 5 B. Energy must be supplied to separate bonded atoms.
- 6 B. More stable bonded arrangements release energy.
- 7 C. $500 - 650 = -150$ kJ.
- 8 B. Insulation and a lid reduce energy transfer.
- 9 B. Not all released energy reaches the water.
- 10 C. Hydrogen combines with oxygen to form water.

Solutions · Structured questions

11. 11(a) Exothermic [1]. (b) From reacting system to surroundings [2]. (c) Products have lower chemical energy [2]. (d) Combustion/neutralisation/respiration and valid observation [3].
12. 12(a) Reactant level above product level [2]. (b) Minimum energy colliding particles need to react [2]. (c) Lower peak / alternative lower-E_a route [2]. (d) Reactant and product energies / Delta H / equilibrium position [any 2].
13. 13(a) $436 + 193 = 629$ kJ/mol [2]. (b) $2 \times 366 = 732$ kJ/mol [2]. (c) $629 - 732 = -103$ kJ/mol [2]. (d) Exothermic because bond making releases more than bond breaking absorbs [2].
14. 14(a) Water temperature change and fuel mass burned [2]. (b) Same water mass/volume, distance, apparatus, initial temperature [any 2]. (c) Lid and insulation/draught shield [any 2]. (d) Less energy released per mass and soot may form [2].
15. 15(a) Carbon dioxide and water [2]. (b) Water [1]. (c) No CO₂ at point of use and high energy per mass / fuel-cell efficiency [2]. (d) Storage/transport/production impacts plus evidence-based conclusion [3].

Self-assessment

Circle the first incorrect step and classify it as knowledge, equation or formula, mole ratio, unit conversion, observation, practical method or communication.

Evidence used: Cambridge IGCSE Co-ordinated Sciences 0654 syllabus 2025-2027 and June 2026 papers, mark schemes and Principal Examiner Report supplied by the user. All practice questions are newly written.